

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Ashburn, Virginia -- station KIAD, America/New_York
Committed pair OSM 744496750 -> 744496741
Amazon Web Services IAD116 / IAD117
Facade gap 60.3 m between the two halls
Weather record 43,763 real hourly records from KIAD
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3550 C at 255 deg

FortyGuard data: this site's own measured FortyGuard forecast/outcome day pairs

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-06-21 (safe_sector)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. That is 2 more than the reactive incumbent, at 0 unsafe hours against its 0.

Agent 2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 0 of 24 hours free cooling, 0 mode change(s), 0 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	20.720	18.0	20.560	0.989
01	mechanical	no	dry-bulb	20.702	18.0	20.560	0.921
02	mechanical	no	dry-bulb	20.685	18.0	20.000	1.090
03	mechanical	no	dry-bulb	20.636	18.0	20.000	0.956
04	mechanical	no	dry-bulb	20.449	18.0	19.440	0.930
05	mechanical	no	dry-bulb	20.427	18.0	19.440	1.109
06	mechanical	no	dry-bulb	20.103	18.0	18.890	0.883
07	mechanical	no	dry-bulb	20.932	18.0	18.330	1.541
08	mechanical	no	dry-bulb	22.087	18.0	17.780	1.921
09	mechanical	no	dry-bulb	22.906	18.0	17.780	2.279
10	mechanical	no	dry-bulb	22.573	18.0	17.780	2.139
11	mechanical	no	dry-bulb	21.193	18.0	17.220	1.687
12	mechanical	no	dry-bulb	20.637	18.0	17.780	1.250
13	mechanical	no	air quality	20.122	18.0	17.780	1.111
14	mechanical	no	air quality	19.601	18.0	17.780	1.224
15	mechanical	no	dew point	19.489	18.0	17.780	1.088
16	mechanical	no	dew point	18.687	18.0	17.780	1.026
17	mechanical	no	dew point	18.126	18.0	17.220	1.295
18	mechanical	no	dew point	17.289	18.0	16.670	1.200
19	mechanical	no	dew point	17.029	18.0	16.670	1.303
20	mechanical	no	dew point	16.489	18.0	16.110	1.467
21	mechanical	no	dew point	16.177	18.0	15.560	1.589
22	FREE	yes	-	16.198	18.0	15.560	1.272
23	FREE	yes	-	15.643	18.0	15.000	1.042

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.720 C against a 18.0 C limit, so it fails by 2.720 C.
It would take a limit 2.720 C higher, or a bound 2.720 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.702 C against a 18.0 C limit, so it fails by 2.702 C.
It would take a limit 2.702 C higher, or a bound 2.702 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.685 C against a 18.0 C limit, so it fails by 2.685 C.
It would take a limit 2.685 C higher, or a bound 2.685 C tighter, to change this hour.

03:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.636 C against a 18.0 C limit, so it fails by 2.636 C.
It would take a limit 2.636 C higher, or a bound 2.636 C tighter, to change this hour.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.449 C against a 18.0 C limit, so it fails by 2.449 C.
It would take a limit 2.449 C higher, or a bound 2.449 C tighter, to change this hour.

05:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.427 C against a 18.0 C limit, so it fails by 2.427 C.
It would take a limit 2.427 C higher, or a bound 2.427 C tighter, to change this hour.

06:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.103 C against a 18.0 C limit, so it fails by 2.103 C.
It would take a limit 2.103 C higher, or a bound 2.103 C tighter, to change this hour.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.932 C against a 18.0 C limit, so it fails by 2.932 C.
It would take a limit 2.932 C higher, or a bound 2.932 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.087 C against a 18.0 C limit, so it fails by 4.087 C.
It would take a limit 4.087 C higher, or a bound 4.087 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.906 C against a 18.0 C limit, so it fails by 4.906 C.
It would take a limit 4.906 C higher, or a bound 4.906 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.573 C against a 18.0 C limit, so it fails by 4.573 C.
It would take a limit 4.573 C higher, or a bound 4.573 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.193 C against a 18.0 C limit, so it fails by 3.193 C.
It would take a limit 3.193 C higher, or a bound 3.193 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.637 C against a 18.0 C limit, so it fails by 2.637 C.
It would take a limit 2.637 C higher, or a bound 2.637 C tighter, to change this hour.

13:00 MECHANICAL [air quality]

Mechanical, and neither temperature nor humidity is the reason. The outside air is too DIRTY: PM2.5 index 75.7 against a 73.5 limit. Opening a damper pulls that air into the hall, which is the documented reason operators avoid free cooling at all.

14:00 MECHANICAL [air quality]

Mechanical, and neither temperature nor humidity is the reason. The outside air is too DIRTY: PM2.5 index 75.3 against a 73.5 limit. Opening a damper pulls that air into the hall, which is the documented reason operators avoid free cooling at all.

15:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 19.489 C would have passed.
The outside air is too HUMID: the dew-point bound is 16.66 C against a 15.0 C maximum, failing by 1.66 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

16:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.687 C would have passed.
The outside air is too HUMID: the dew-point bound is 16.67 C against a 15.0 C maximum, failing by

1.67 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

17:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.126 C would have passed. The outside air is too HUMID: the dew-point bound is 16.39 C against a 15.0 C maximum, failing by 1.39 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

18:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.289 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

19:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.029 C would have passed. The outside air is too HUMID: the dew-point bound is 16.11 C against a 15.0 C maximum, failing by 1.11 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.489 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.177 C would have passed. The outside air is too HUMID: the dew-point bound is 15.55 C against a 15.0 C maximum, failing by 0.55 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.198 C, which is 1.802 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.272 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.0827 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.643 C, which is 2.357 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.042 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.0885 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

WHAT IT IS WORTH OVER FIVE REAL YEARS, AT THIS SITE

Held-out days simulated 913 -- the agent never calibrates on a day it is scored on
Free cooling delivered 14.72 h/day, i.e. 5,375 h/yr
Chiller-hours avoided +405.7 h/yr against a tuned reactive incumbent
12-hour plan stability 94.1 % of 21,879 re-plans change nothing at all

The ladder, one constraint at a time:

N-56-like: notice 0, skill 1.00, no constraint +65.6 h/yr
+ switch budget 2, min dwell 3 h +85.6 h/yr
+ dew-point gate 15 C (Green Grid WP#46 p.6) +118.8 h/yr
+ notice 3 h, skill 0.50 (no perfect forecast) +405.7 h/yr
+ unanchored, 4 measured FG offsets rotated -156.0 h/yr

PRICED, IN THIS STATE'S OWN ELECTRICITY TARIFF

Chiller power 156.1 kW per MW of IT load (0.549 kW/ton, the ASHRAE 90.1-2019 minimum)
Electricity 8.72 cents/kWh (Virginia commercial, 2024 annual)
Energy avoided 63,325 kWh per MW of IT load per year
Value \$5,522 per MW of IT load per year

Everything above is PER MEGAWATT OF IT LOAD. This project has never measured a data centre's size and will not invent one; a reader who knows their own IT load multiplies once.

WHAT IS NOT CLAIMED

- The chiller COMPRESSOR only. Fans, chilled-water pumps, condenser pumps and cooling-tower fans keep running, and an airside economizer moves MORE air, so fan power can rise. The unmeasured term has the OPPOSITE SIGN, so this is an upper bound.
- Code-minimum chiller efficiency is the OPTIMISTIC end: Standard 90.1 is a floor, hyperscale plants beat it, and a better chiller saves less per hour switched off.
- Full-load kW/ton OVERSTATES the draw at the moment free cooling is available, because free cooling happens when it is cool and a chiller at a cool condenser runs below its rating. IPLV is the lower published number but is an annual-weighted metric under AHRI's assumed load profile, not the part-load point free-cooling weather sits at.
- The heat rejected is APPROXIMATED by the IT load. UPS, PDU and lighting losses inside the envelope add to it; some overhead sits outside the cooled space, so scaling by full PUE would overshoot. No PUE multiplier is applied.
- EIA STATE-SECTOR AVERAGES, not the site's tariff. A Loudoun County campus buys on a large-general-service contract that is not public.
- No carbon, no water, no maintenance and no demand charges. Tower water use moves the OTHER way when the chiller runs less.
- Every caveat on the hours is inherited: the headline is conditional on the bank sitting on the long facade (the refusal guard is priced at -3,124 h/yr where it fires) and the unanchored row is NEGATIVE. Those rows appear here with their signs intact.

HOW TO REPRODUCE EVERY NUMBER IN THIS REPORT

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cd INTAKE-ARBITER/src && python run_all.py
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That rebuilds every artefact from saved data and then audits it: 39 checks, 68 published figures re-read out of the files the code itself wrote, and it exits non-zero on any failure. It makes ZERO API calls -- every input is a saved FortyGuard response, a committed geometry file, or the station's own hourly record.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.